Lectures 5-6: Laplace Transforms and Properties - Part 1

CBE 162, Fall 2026

We will introduce two new topics: (i) deviation variables and (ii) Laplace transforms. Once we learn how Laplace transforms are defined and the key superposition property, we will discuss how they can be used for analytical solution of linear ODEs.

1 Deviation Variables

Deviation variables are a useful tool for understanding the dynamics of a process relative to a given steady-state value. For any given dynamic variable $x(t)$ (could be CV, MV, or DV), we define its deviation variable form as

$$\tilde{x}(t) = x(t) - x_{ss}, \tag{1}$$

where x_{ss} denotes the steady-state value of x before any disturbance (DV) or manipulated (MV) variable changes have occurred in the system. Note that sometimes the overbar notation $\bar{x}$ is used to denote steady-state values; here we will use the subscript x_{ss}. The main advantage of x_{ss} versus $\bar{x}$ is so that the tilde accent $\tilde{x}$ does not get confused with the overbar.

Why use deviation variables? Although deviation variables simply shift the original variable by a constant, we can ensure that $\tilde{x}(t) = x(0) - x_{ss} = 0$ whenever $x(0) = x_{ss}$, as we will typically assume before a disturbance upsets the process. This allows us to avoid the initial conditions effects. Furthermore, since we can easily convert our ordinary differential equations (ODEs) to be in terms of $\tilde{x}(t)$, we can simply solve for $\tilde{x}(t)$ and then convert back to $x(t) = x_{ss} + \tilde{x}(t)$ once we have the solution (if needed).

Note: At first, we will be explicit about denoting deviation variables with a $\tilde{\cdot}$; however, later on in the class we will stop writing the tildes on everything, so you will need to be able to infer from context if something is in deviation variable form or not.

Example: Heated stirred tank

Let us see how deviation variables help us in the context of an example. We will use the simplified form of the heated stirred tank energy balance derived before:

$$\tau \frac{dT}{dt} + T = T_i + K_p Q. \tag{2}$$

To solve such an equation, we are missing a crucial component, which is the *initial condition* for the "state" variable (i.e., the variable appearing in the derivative, which is T in this case). Although we could initialize at any time point t_0, we often take $t_0 = 0$ for simplicity, such that we need to specify $T(0)$. You would need to be told this value in a given application (or infer it from the problem statement or other problem data). Typically, we assume that the process start at some steady-state value $T(0) = T_{ss}$, as we want to explore what happens when an upset occurs for some $t \geq 0$. We know that any steady-state should satisfy (2) with the time derivative term set to zero, i.e.,

$$T_{ss} = T_{i,ss} + K_p Q_{ss}. \tag{3}$$

It is important to recognize that we do not need to find a specific value for T_{ss}; the subsequent analysis will hold for any value that satisfies this equation, which clearly depends on the specific choice of the steady-state DV $T_{i,ss}$ and MV Q_{ss} values.

Derivation of ODE in terms of deviation variables

Let us now walk through the derivation of a new ODE that is defined in terms of deviation variables:

Start by substracting T_{ss} from both sides of (2)

$$\tau\frac{dT}{dt} + T - T_{ss} = T_i + K_pQ - T_{ss}, \qquad T(0) = T_{ss}, \tag{4a}$$

Substitute the steady-state expression (3) in the right-hand side

$$\tau\frac{dT}{dt} + T - T_{ss} = T_i + K_pQ - (T_{i,ss} + K_pQ_{ss}), \qquad T(0) = T_{ss}, \tag{4b}$$

Group terms so that they look like deivation variables

$$\tau\frac{dT}{dt} + (T - T_{ss}) = (T_i - T_{i,ss}) + K_p\left(Q - Q_{ss}\right), \qquad T(0) = T_{ss}, \tag{4c}$$

Derivative of a constant is zero $\frac{dT}{dt} = \frac{d(\tilde{T}+T_{ss})}{dt} = \frac{d\tilde{T}}{dt}$

$$\tau\frac{d\tilde{T}}{dt} + \tilde{T} = \tilde{T}_i + K_p\tilde{Q}, \qquad \tilde{T}(0) = 0, \tag{4d}$$

We see that the final expression is defined fully in terms of the deviation variables for the CV, MV, and DV.

Solving ODE in deviation form

We now want to solve (4d). DOF analysis tells us that we must specify functional forms for the DV and MV in order to have a solvable equation. Let us assume that T_i remains constant at its steady-state value and Q undergoes a step change. An illustration of the step change is given in Figure 1.

Question: What is the temperature profile $\tilde{T}(t)$ when $T_i = T_{ss}$ remains constant and Q undergoes the step change in Figure 1?

Answer: To start, we notice the deviation variables already provide us an advantage when dealing with the inputs, i.e., $\tilde{T}_i = 0$ and $\tilde{Q} = \Delta Q$ for all $t \geq 0$. Thus, we can immediately reduce (4d) to $\tau\frac{d\tilde{T}}{dt} + \tilde{T} = K_p\Delta Q$ with $\tilde{T}(0) = 0$. There are many ways to solve this differential equation – since it is separable we provide one such derivation in Figure 2. The final answer that we obtain is

$$\tilde{T}(t) = K_p\Delta Q(1 - \exp(-t/\tau)). \tag{5}$$

Note that, during the deviation, we are able to use the fact that $\tilde{T}(0) = 0$ to remove some terms, meaning that the deviation variables helped simplify the resulting expression in multiple ways.

Let us now interpret (5) in more detail. Clearly, $\tilde{T}(t)$ starts from 0 at $t = 0$ and goes to a limiting value of $K_p\Delta Q$ as $t \to \infty$. Letting $\Delta T = (T(\infty) - T_{ss}) = \tilde{T}(\infty)$ denote the total change in the temperature, we see that $K_p = \frac{\Delta T}{\Delta Q}$, i.e., the change in the CV divided by the change in the MV. Thus, the parameter K_p has an important meaning in that it tells us how the CV will change for a unit change in the MV. This is often referred to as the "process gain" and we will learn more about this in future lectures. The parameter τ, on the other hand, does not affect the limiting value of the CV; however, it does affect the *speed* at which $T(t)$ reaches its new steady-state value. Larger values of τ imply the exponential term requires much larger

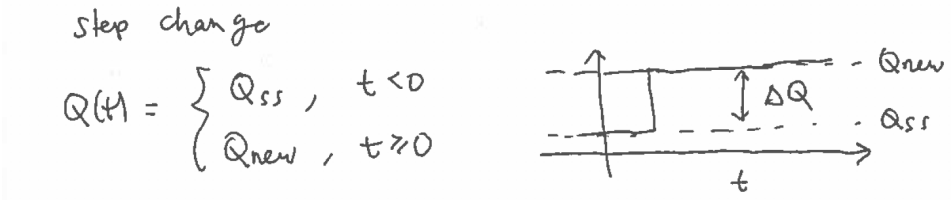


Figure 1: Illustration of step change in heating rate $Q(t)$. The step occurs at time $t = 0$.

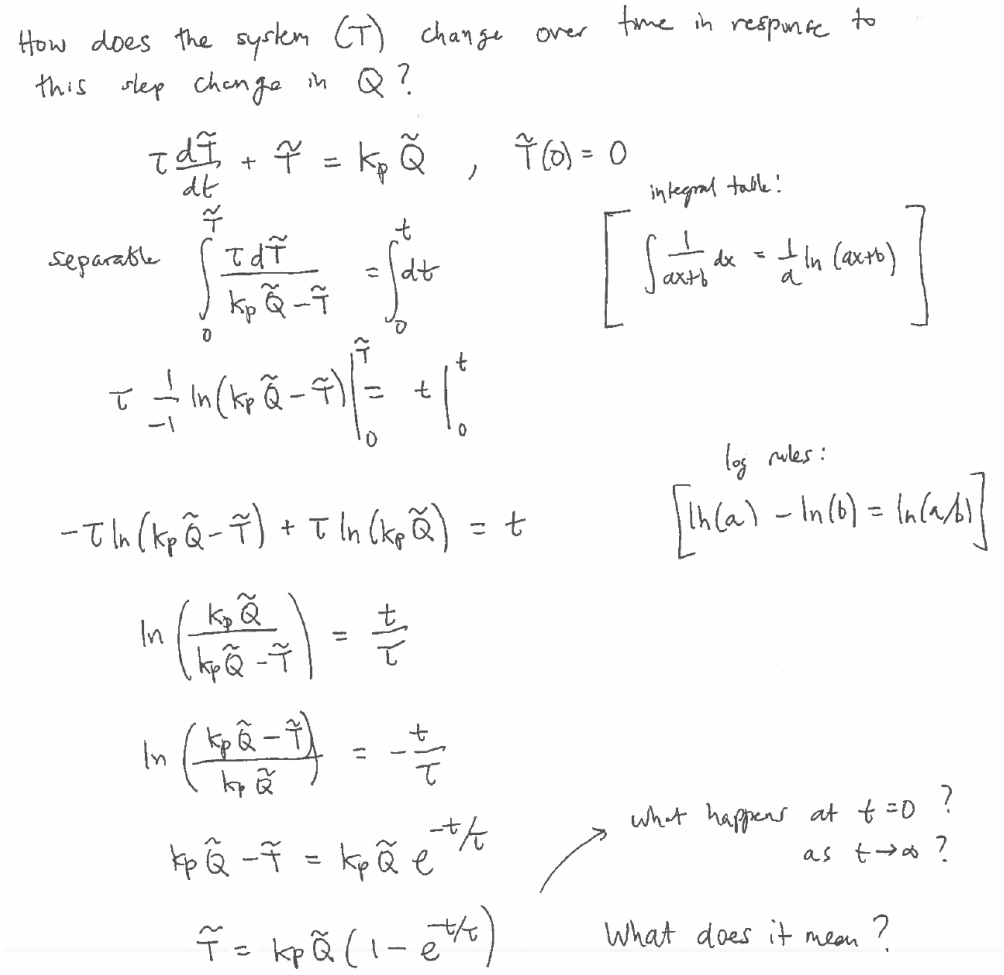


Figure 2: Derivation of analytic solution to temperature profile (in deviation form).

t values to decrease such that the system response is much slower overall. As such, the parameter τ is often referred to as the "time constant."

2 Laplace Transforms

We now introduce the notion of *Laplace transforms* (sometimes referred to as LT for short), which will be heavily used throughout the course.

2.1 Why Laplace Transforms?

Dynamic models result in ODEs that we must solve either numerically or analytically. Analytic solutions (when possible) are valuable in that we can see how changes in certain parameters directly affect the solution without the need for specialized software. In the example above, we solved a first-order ODE in the time domain. This process becomes cumbersome as the ODE gets more complicated (second- or higher-order).

Pierre Simon Marquis de Laplace (a famous French mathematician) developed an integral transformation that transforms the ODE to a new space (often called the s-domain as opposed to the time domain) that simplifies the process for solving *linear* ODEs. The main idea is to convert differentials in time t to algebraic expressions in terms of the new variable s. *Thus, the main purpose of the Laplace transform is to simplify the process of understanding, manipulating, and solving dynamic system models.*

2.2 Formal Definition

2.2.1 Forward transform

The **Laplace transform** is defined as follows

$$\mathscr{L}[f(t)] := F(s) = \int_{0^-}^{\infty} f(t)e^{-st}dt, \tag{6}$$

where 0^- means the value of 0 evaluated from the negative direction ($t \to 0$ from the left). The reason we need to use 0^- above is that we want to capture impulses that happen directly at time $t = 0$.

The Laplace transform takes a function $f(t)$ in the "time-domain" and converts is to the "s-domain" denoted by $F(s)$. In this case, s is a complex variable that can be expressed in Cartesian form as

$$s = a + bj \tag{7}$$

where $a = \text{Re}(s)$ is the real part of s and $b = \text{Im}(s)$ is the imaginary part of s; both a and b are real numbers in this form and j denotes the imaginary unit with $j^2 = -1$. We can also express complex numbers in polar form $s = re^{\phi j}$ where $r = \sqrt{a^2 + b^2}$ and $\phi = \tan^{-1}(b/a)$.

Note: We use j to denote the imaginary unit, but several books/references use i instead. You can use whatever notation you are comfortable with in this class, as long as you are clear.

2.2.2 Inverse transform

The Laplace transform operator $\mathscr{L}[\cdot]$ is only defined for functions with a single output. Thus, we can define the corresponding *unique* inverse transform as

$$f(t) = \mathscr{L}^{-1}[F(s)], \quad \text{for all } t \geq 0. \tag{8}$$

Note that the inverse transform $\mathscr{L}^{-1}[\cdot]$ is only defined for $t \geq 0$, so it contains no information about $f(t)$ for any $t < 0$. Although there is an explicit formula for the inverse Laplace transform (known as Mellin's inverse formula), it is difficult to evaluate as it involves the limit of a line integral over the imaginary part of s. Therefore, in practice, we typically determine the inverse Laplace transform by inspection of known forward transformations that lead to $F(s)$, i.e., $f(t) \Leftrightarrow F(s)$, which must be unique. These transformations are typically summarized in table form. Let us highlight a key property of Laplace transforms and demonstrate how (6) can be used.

2.2.3 The Principle of Superposition

The Laplace transform is a *linear operator* since it satisfies the following principle of superposition for any constants a and b

$$\mathscr{L}\left(ax(t) + by(t)\right) = a\mathscr{L}(x(t)) + b\mathscr{L}(y(t)) = aX(s) + bY(s). \tag{9}$$

This is the key property that will allow us to solve challenging linear ODEs in much simpler fashion.

2.3 Examples from Scratch

Although we will often use tables to get Laplace transforms and their inverses, let us see how these tables were populated by applying (6) to some common example functions and operators.

2.3.1 Example: Step Function

A unit step function can be written out as follows

$$\gamma(t) = \begin{cases} 0 & \text{if } t < 0 \\ 1 & \text{if } t \geq 0 \end{cases} \tag{10}$$

Note that this function is discontinuous at $t = 0$.

Laplace transform of unit step function:

Start with definition (6)

$$\mathscr{L}[\gamma(t)] = \int_{0^-}^{\infty} \gamma(t) e^{-st} dt, \tag{11a}$$

We can break integral up into different ranges

$$= \underbrace{\int_{0^-}^{0+} \gamma(t) e^{-st} dt}_{=0 \text{ since step happens to the right}} + \int_{0^+}^{\infty} \underbrace{\gamma(t)}_{=1 \text{ in this range}} e^{-st} dt = \int_{0^+}^{\infty} e^{-st} dt, \tag{11b}$$

Recall integral of exponential

$$= -\frac{1}{s} e^{-st}\Big|_{0^+}^{\infty} = -\frac{1}{s}\left(e^{-\infty} - e^{-0^+}\right) = -\frac{1}{s}(0 - 1), \tag{11c}$$

Simplify the expression

$$\mathscr{L}[\gamma(t)] = \frac{1}{s}. \tag{11d}$$

Question: Can we directly work with the Laplace transform of a constant function $f(t) = c$? No, since Laplace transformation is only defined for $t \geq 0$, so all functions are implicitly assumed to be zero before $t = 0$. For the purposes of this class, $f(t) = c\gamma(t) \to F(s) = \frac{c}{s}$ where $\gamma(t)$ denotes the step function.

2.3.2 Example: Derivative Operator

Let us see what happens when we apply the Laplace transform to the derivative operator $\frac{df}{dt}$.

Laplace transform of derivative operator:

Start with definition (6)

$$\mathscr{L}\left[\frac{df}{dt}\right] = \int_{0^-}^{\infty} \frac{df}{dt} e^{-st} dt. \tag{12a}$$

Recall integration by parts $\int u dv = uv - \int v du$. We can select $dv = \frac{df}{dt} dt$ and $u = e^{-st}$ such that $v = \int dv = \int \frac{df}{dt} dt = f(t)$ and $du = -se^{-st} dt$

$$= \left[e^{-st} f(t)\right]_{0^-}^{\infty} + \int_{0^-}^{\infty} f(t)(-s) e^{-st} dt, \tag{12b}$$

Simplify the first term

$$= \underbrace{e^{-\infty}}_{0} f(\infty) - \underbrace{e^{-0^-}}_{1} f(0^-) - \int_{0^-}^{\infty} f(t)(-s) e^{-st} dt = \int_{0^-}^{\infty} f(t) s e^{-st} dt - f(0^-), \tag{12c}$$

We can pull s out of the second term

$$= s \int_{0^-}^{\infty} f(t) e^{-st} dt - f(0^-), \tag{12d}$$

Notice that the first term now has the exact same form as the Laplace transform definition in (6)

$$\mathscr{L}\left[\frac{df}{dt}\right] = sF(s) - f(0^-). \tag{12e}$$

We can see that the initial condition of the time-domain function $f(0^-)$ appears in this expression; remember that this is evaluated from the negative direction. What would this be in deviation variables assuming the system started from steady state? $\tilde{f}(0^-) = f(0^-) - f_{ss} = f_{ss} - f_{ss} = 0$.

2.3.3 Example: Exponential Function

As a final example, let us look at the exponential function e^{-at} for some constant a.

Laplace transform of exponential function:

Start with definition (6)

$$\mathscr{L}\left[e^{-at}\right] = \int_{0^-}^{\infty} e^{-at} e^{-st} dt, \tag{13a}$$

Combine the exponential terms

$$= \int_{0^-}^{\infty} e^{-(s+a)t} dt, \tag{13b}$$

Recall integral of exponential

$$= \left[-\frac{1}{s+a} e^{-(s+a)t}\right]_{0^-}^{\infty} = -\frac{1}{s+a} e^{-\infty} + \frac{1}{s+a} e^{0^-}, \tag{13c}$$

Simplify expression

$$\mathscr{L}\left[e^{-at}\right] = \frac{1}{s+a}. \tag{13d}$$

Note: By convention, all time domain functions are implicitly defined as being 0 before time 0 to ensure the Laplace transform is unique. This can be stated mathematically as multiplying any function $f(t)$ by a step function, i.e., $f(t)\gamma(t)$. For example, the exponential function e^{-at} above should really be stated as $e^{-at}\gamma(t)$. To simplify notation, we often drop the step function when writing out derivations. A footnote about this point is provided at the bottom of the Laplace transform table (Figure 3).

3 Laplace Transform Tables

As opposed to working out the Laplace transforms by hand, it is convenient to store them in a table; for example, see Figures 3 and 4. Figure 3 shows common Laplace transform pairs $f(t) \Leftrightarrow F(s)$, e.g., $e^{-at} \Leftrightarrow \frac{1}{s+a}$. Figure 4 shows important properties of Laplace transforms including how different operators transform functions in the time- and s-domain.

Example: Heated stirred tank

We will learn about a general procedure for solving ODEs using Laplace transforms later. Let us first apply the process to the heated stirred tank example that we already solved in (5) to see if we can get the same answer more efficiently. We start with the following ODE in deviation variable form

$$\tau \frac{d\tilde{T}}{dt} + \tilde{T} = \tilde{T}_i + K_p \tilde{Q}, \qquad \tilde{T}(0) = 0, \tag{14}$$

and again consider $\tilde{T}_i = 0$ and $\tilde{Q}(t) = \Delta Q \gamma(t)$ (step of size ΔQ).

Laplace transform applied to heated stirred tank ODE:

Start by substituting the DV and MV expressions

$$\tau \frac{d\tilde{T}}{dt} + \tilde{T} = K_p \Delta Q \gamma(t), \tag{15a}$$

Apply the Laplace transform operator to both sides

$$\mathscr{L}\left[\tau \frac{d\tilde{T}}{dt} + \tilde{T}\right] = \mathscr{L}\left[K_p \Delta Q \gamma(t)\right], \tag{15b}$$

Use linearity of $\mathscr{L}[\cdot]$ to simplify

$$\tau \mathscr{L}\left[\frac{d\tilde{T}}{dt}\right] + \mathscr{L}\left[\tilde{T}\right] = K_p \Delta Q \mathscr{L}\left[\gamma(t)\right], \tag{15c}$$

From the table (or previous expressions we derived), we can simplify

$$\tau\left(s\tilde{T}(s)-\underbrace{\tilde{T}(0^-)}_{=0}\right)+\tilde{T}(s)=K_p\Delta Q\frac{1}{s}, \tag{15d}$$

Manipulate in the s-domain

$$(\tau s+1)\tilde{T}(s)=K_p\Delta Q\frac{1}{s}\quad\rightarrow\quad\tilde{T}(s)=\frac{K_p\Delta Q}{s(\tau s+1)}, \tag{15e}$$

Apply inverse transform to get back to time domain

$$\tilde{T}(t)=\mathscr{L}^{-1}\left[\tilde{T}(s)\right]=\mathscr{L}^{-1}\left[\frac{K_p\Delta Q}{s(\tau s+1)}\right]=\frac{K_p\Delta Q}{\tau}\mathscr{L}^{-1}\left[\frac{1}{s\left(s+\frac{1}{\tau}\right)}\right], \tag{15f}$$

See entry #5 in Figure 3

$$\tilde{T}(t)=\frac{K_p\Delta Q}{\tau}\frac{1}{\frac{1}{\tau}}\left(1-\exp(-t/\tau)\right)=K_p\Delta Q\left(1-\exp(-t/\tau)\right). \tag{15g}$$

The final answers match!

Common Laplace Transform Pairs

Time Domain Function		Laplace Domain Function
Name	Definition*	
Unit Impulse	$\delta(t)$	1
Unit Step	$\gamma(t)$ †	$\frac{1}{s}$
Unit Ramp	t	$\frac{1}{s^2}$
Parabola	t^2	$\frac{2}{s^3}$
Exponential	e^{-at}	$\frac{1}{s+a}$
Asymptotic Exponential	$\frac{1}{a}(1-e^{-at})$	$\frac{1}{s(s+a)}$
Dual Exponential	$\frac{1}{b-a}\left(e^{-at}-e^{-bt}\right)$	$\frac{1}{(s+a)(s+b)}$
Asymptotic Dual Exponential	$\frac{1}{ab}\left[1+\frac{1}{a-b}\left(be^{-at}-ae^{-bt}\right)\right]$	$\frac{1}{s(s+a)(s+b)}$
Time multiplied Exponential	te^{-at}	$\frac{1}{(s+a)^2}$
Sine	$\sin(\omega_0 t)$	$\frac{\omega_0}{s^2+\omega_0^2}$
Cosine	$\cos(\omega_0 t)$	$\frac{s}{s^2+\omega_0^2}$
Decaying Sine	$e^{-at}\sin(\omega_d t)$	$\frac{\omega_d}{(s+a)^2+\omega_d^2}$
Decaying Cosine	$e^{-at}\cos(\omega_d t)$	$\frac{s+a}{(s+a)^2+\omega_d^2}$
Generic Oscillatory Decay	$e^{-at}\left[B\cos(\omega_d t)+\frac{C-aB}{\omega_d}\sin(\omega_d t)\right]$	$\frac{Bs+C}{(s+a)^2+\omega_d^2}$
Prototype Second Order Lowpass, underdamped	$\frac{\omega_0}{\sqrt{1-\zeta^2}}e^{-\zeta\omega_0 t}\sin\left(\omega_0\sqrt{1-\zeta^2}t\right)$	$\frac{\omega_0^2}{s^2+2\zeta\omega_0 s+\omega_0^2}$
Prototype Second Order Lowpass, underdamped - Step Response	$1-\frac{1}{\sqrt{1-\zeta^2}}e^{-\zeta\omega_0 t}\sin\left(\omega_0\sqrt{1-\zeta^2}t+\phi\right)$ $\phi=\tan^{-1}\left(\frac{\sqrt{1-\zeta^2}}{\zeta}\right)$	$\frac{\omega_0^2}{s(s^2+2\zeta\omega_0 s+\omega_0^2)}$

*All time domain functions are implicitly=0 for t<0 (i.e. they are multiplied by unit step, γ(t)).
†u(t) is more commonly used for the step, but is also used for other things. γ(t) is chosen to avoid confusion (and because in the Laplace domain it looks a little like a step function, Γ(s)).

Figure 3: Common Laplace transform pairs.

Common Laplace Transform Properties

Name	Illustration
Definition of Transform	$\mathrm{f(t)} \overset{\mathrm{L}}{\longleftrightarrow} \mathrm{F(s)}$ $\mathrm{F(s)} = \int_{0^-}^{\infty} \mathrm{f(t)e^{-st}dt}$
Linearity	$Af_1(t) + Bf_2(t) \overset{L}{\longleftrightarrow} AF_1(s) + BF_2(s)$
First Derivative	$\frac{df(t)}{dt} \overset{L}{\longleftrightarrow} sF(s) - f(0^-)$
Second Derivative	$\frac{d^2 f(t)}{dt^2} \overset{L}{\longleftrightarrow} s^2 F(s) - sf(0^-) - \dot{f}(0^-)$
nth Derivative	$\frac{d^n f(t)}{dt^n} \overset{L}{\longleftrightarrow} s^n F(s) - \sum_{i=1}^{n} s^{n-i} f^{(i-1)}(0^-)$
Integral	$\int_0^t f(\lambda)d\lambda \overset{L}{\longleftrightarrow} \frac{1}{s}F(s)$
Time Multiplication	$tf(t) \overset{L}{\longleftrightarrow} -\frac{dF(s)}{ds}$
Time Delay	$\mathrm{f(t-a)\gamma(t-a)} \overset{\mathrm{L}}{\longleftrightarrow} \mathrm{e^{-as}F(s)}$ γ(t) is unit step
Complex Shift	$f(t)e^{-at} \overset{L}{\longleftrightarrow} F(s+a)$
Scaling	$f\left(\frac{t}{a}\right) \overset{L}{\longleftrightarrow} aF(as)$
Convolution Property	$f_1(t) * f_2(t) \overset{L}{\longleftrightarrow} F_1(s)F_2(s)$
Initial Value	$\lim_{t\to 0^+} f(t) = \lim_{s\to\infty} sF(s)$
Final Value (if final value exists)	$\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$

Figure 4: Common Laplace transform properties.